

4_10xMouse_PBMC_comparison_geneIntersection

August 31, 2026

```
[1]: library(rtracklayer)
library(ggplot2)
library(Seurat)

getwd()
dir.create("figures_10xMouse_PBMC")
dir.create("data")
colorDict = c("2CLC"="#88535A",
              "Zscan4+"="#EF8264",
              "Pluri"="#F2CC8F")

colorAge = c("old"="#A58065", #A67C63, #DB9D74
            "young"="#7FCFF2") #049DD9

colorTools = c("SoloTE"="#A4DD9B", #6fc69d",
              "Stellarscope"="#4f5d93",
              "STARsolo"="#f0df93")

dataset_id <- "10xMouse_RA"
```

Loading required package: GenomicRanges

Loading required package: stats4

Loading required package: BiocGenerics

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,

colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
Position, rank, rbind, Reduce, rownames, sapply, setdiff, sort,
table, tapply, union, unique, unsplit, which.max, which.min

Loading required package: S4Vectors

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomeInfoDb

Loading required package: SeuratObject

Loading required package: sp

Attaching package: 'sp'

The following object is masked from 'package:IRanges':

%over%

Attaching package: 'SeuratObject'

The following object is masked from 'package:GenomicRanges':

intersect

The following object is masked from 'package:GenomeInfoDb':

```
intersect
```

The following object is masked from 'package:IRanges':

```
intersect
```

The following object is masked from 'package:S4Vectors':

```
intersect
```

The following object is masked from 'package:BiocGenerics':

```
intersect
```

The following objects are masked from 'package:base':

```
intersect, t
```

```
~/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xMouse_PBMC"):
```

```
"'figures_10xMouse_PBMC' already exists"
```

```
Warning message in dir.create("data"):
```

```
"'data' already exists"
```

```
[2]: # theme_paper <- function(base_size = 17, base_family = "") {  
#   theme_minimal(base_size = base_size, base_family = base_family) +  
#     theme(  
#       plot.title = element_text(face = "bold", size = base_size + 2, hjust = 0.5),  
#       axis.title = element_text(size = base_size),  
#       axis.text = element_text(size = base_size * 0.9),  
#       legend.title = element_text(size = base_size),  
#       legend.text = element_text(size = base_size * 0.9),  
#       panel.grid.major = element_line(linewidth = 0.4),  
#       panel.grid.minor = element_blank(),  
#       plot.margin = margin(10, 10, 10, 10)  
#     )  
# }
```

```
# theme_set(theme_paper())
```

```
[2]: #save.image("workspaces/afterCorrelations_wSTAR.RData")
load("workspaces/afterCorrelations_wSTAR_thr5percCells_10xMouse_PBMC.RData")
rm(mat_stellarscope)
rm(mat_soloTE)
rm(mat_STARsolo)
gc()
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	28207958	1506.5	51845261	2768.9	28221813	1507.3
	Vcells	259429809	1979.3	386726289	2950.5	273754817	2088.6

1 Gene Intersections

```
[4]: library(rtracklayer)
# import the annotation used for snakemake
gene_annotation <- rtracklayer::import("/mnt/TEdataStorage_2T/snakemake_data/
↳ references/mm10/genencode.vM10.annotation.gtf.gz")
names(mcols(gene_annotation))
```

1. 'source' 2. 'type' 3. 'score' 4. 'phase' 5. 'gene_id' 6. 'gene_type' 7. 'gene_status'
8. 'gene_name' 9. 'level' 10. 'havana_gene' 11. 'transcript_id' 12. 'transcript_type'
13. 'transcript_status' 14. 'transcript_name' 15. 'transcript_support_level' 16. 'tag' 17. 'ha-
vana_transcript' 18. 'exon_number' 19. 'exon_id' 20. 'protein_id' 21. 'ccdsid' 22. 'ont'

```
[5]: gene_annotation_df <- as.data.frame(gene_annotation)

# Extract only exon entries
exons <- gene_annotation[gene_annotation$type == "exon"]
names(exons) <- exons$gene_name
```

```
[6]: # keep only TEs that are in both SoloTE and Stellarscope
annotation_common <- na.omit(conversionTable)

cat("Classes not in SoloTE annotation: ",
    paste(setdiff(unique(conversionTable$class),
↳ unique(annotation_common$class)), collapse = ", "),
    "\n")
```

Classes not in SoloTE annotation: Other, Satellite, Unknown, NA, RNA

```
[7]: objTElist <- list()

objTElist[["SoloTE"]] <- objTE_soloTE
objTElist[["Stellarscope"]] <- objTE_stellarscope
```

```
objTElist[["STARsolo"]] <- objTE_STARsolo
```

```
objTElist
```

```
$SoloTE
```

```
An object of class Seurat
```

```
5654 features across 5015 samples within 1 assay
```

```
Active assay: RNA (5654 features, 4000 variable features)
```

```
3 layers present: counts, data, scale.data
```

```
2 dimensional reductions calculated: pca, umap
```

```
$Stellarscope
```

```
An object of class Seurat
```

```
4306 features across 5015 samples within 1 assay
```

```
Active assay: RNA (4306 features, 4000 variable features)
```

```
3 layers present: counts, data, scale.data
```

```
2 dimensional reductions calculated: pca, umap
```

```
$STARsolo
```

```
An object of class Seurat
```

```
3247 features across 5015 samples within 1 assay
```

```
Active assay: RNA (3247 features, 3247 variable features)
```

```
3 layers present: counts, data, scale.data
```

```
2 dimensional reductions calculated: pca, umap
```

```
[8]: # Compute overlaps to exons and save lists of loci
featuresList <- list()

bedList <- list()
grList <- list()
exonOverlapList <- list()
toolOverlappingGrList <- list()
toolOverlappingList <- list()

for(tool in names(objTElist)){
  if(tool == "SoloTE"){
    bedList[[tool]] <- conversionTable[conversionTable$soloteID %in%
↳Features(objTElist[[tool]]),
    c("chr","start","end","strand","stellarscopeID")]
    print(nrow(bedList[[tool]]))
  }else{
    bedList[[tool]] <- conversionTable[conversionTable$stellarscopeID %in%
↳Features(objTElist[[tool]]),
    c("chr","start","end","strand","stellarscopeID")]
    print(nrow(bedList[[tool]]))
  }
}
```

```

    bedList[[tool]] <- bedList[[tool]][!
↪duplicated(bedList[[tool]]$stellarscopeID),]
    bedList[[tool]] <- bedList[[tool]][!is.na(bedList[[tool]]$stellarscopeID),]
    featuresList[[tool]] <- bedList[[tool]]$stellarscopeID
    print(nrow(bedList[[tool]]))
    rownames(bedList[[tool]]) <- bedList[[tool]]$stellarscopeID
    # turn into GR object
    grList[[tool]] <- makeGRangesFromDataFrame(bedList[[tool]])
    exonOverlapList[[tool]] <- findOverlaps(grList[[tool]], exons, minoverlap =
↪10)

    toolOverlappingGrList[[tool]] <- grList[[tool]][unique(
↪exonOverlapList[[tool]]@from)]
    toolOverlappingList[[tool]] <- rownames(as.data.
↪frame(toolOverlappingGrList[[tool]]))
    #print(as.data.frame(toolOverlappingList[[tool]]))
}

```

[1] 5649

[1] 5646

[1] 4306

[1] 4306

Warning message in .merge_two_Seqinfo_objects(x, y):

"Each of the 2 combined objects has sequence levels not in the other:

- in 'x': GL456210.1, GL456216.1, GL456221.1, GL456372.1, JH584304.1

- in 'y': chrM

Make sure to always combine/compare objects based on the same reference genome (use suppressWarnings() to suppress this warning)."

[1] 3247

[1] 3247

Warning message in .merge_two_Seqinfo_objects(x, y):

"Each of the 2 combined objects has sequence levels not in the other:

- in 'x': GL456210.1, GL456211.1, GL456221.1, JH584304.1

- in 'y': chrM

Make sure to always combine/compare objects based on the same reference genome (use suppressWarnings() to suppress this warning)."

[9]: colorTools

SoloTE '#A4DD9B' Stellarscope '#4f5d93' STARsolo '#f0df93'

```

[10]: library(UpSetR)
      options(repr.plot.width=9, repr.plot.height=6)

      ph <- UpSetR::upset(fromList(toolOverlappingList),

```

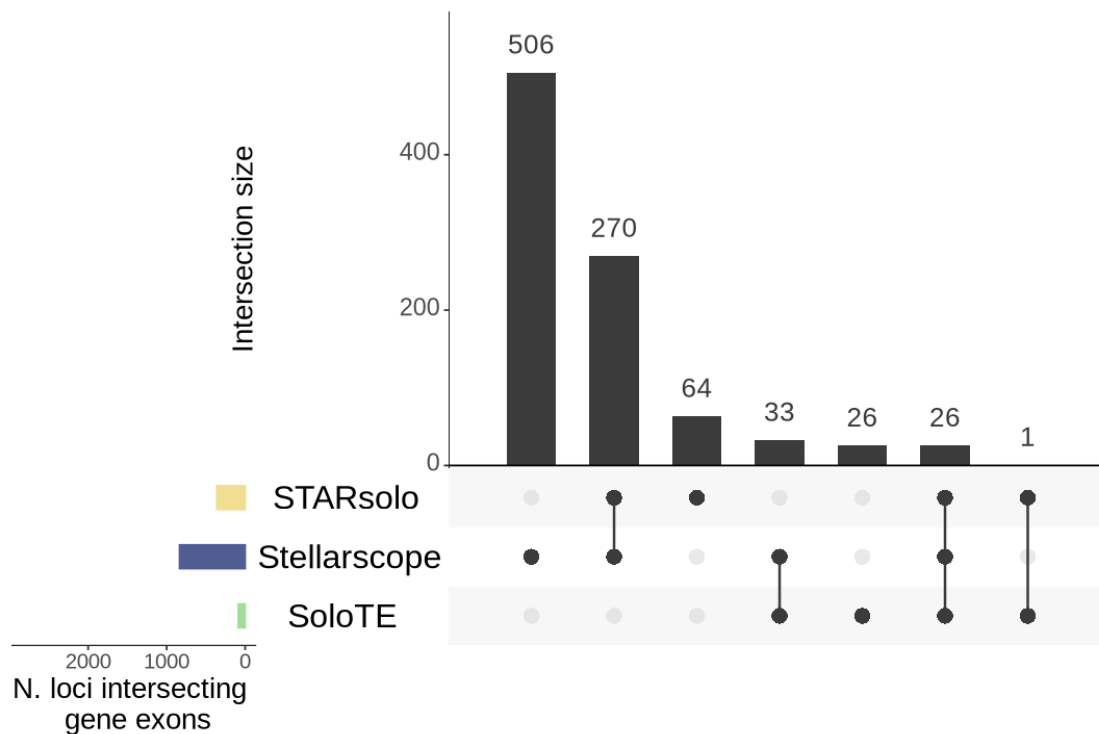
```

sets=names(tool0verlappingList), sets.bar.color=colorTools,
keep.order = TRUE,
text.scale = c(2, 2, 2, 1.75, 2.75, 2.5),
order.by=c("freq"),
point.size=4, mb.ratio = c(0.63, 0.37),
set_size.show=F, set_size.scale_max=
↪max(lengths(tool0verlappingList))+2000,
sets.x.label="N. loci intersecting \n gene exons",
mainbar.y.label="Intersection size")

pdf(paste0("figures_", dataset_id, "/NlociIntersectingGeneExons_upset.pdf"),
↪width=9, height=6)
ph
dev.off()
ph

```

pdf: 2



```

[11]: df <- NULL

for(tool in names(objTElist)){

  loci <- bedList[[tool]]$stellarscopeID

```

```

lociOverlappingGenes <- toolOverlappingList[[tool]]

df <- rbind(df, c(tool, length(loci), length(lociOverlappingGenes)))
}

df <- as.data.frame(df)
df

```

A data.frame: 3 × 3

	V1 <chr>	V2 <chr>	V3 <chr>
	SoloTE	5646	86
	Stellarscope	4306	835
	STARsolo	3247	361

[]:

[]: